

Causal Inference HW3

The purpose of this assignment is to help you understand and apply **mediation analysis** to investigate the mechanisms through which an independent variable affects an outcome via a mediator. We will use the **mtcars** dataset, which is included by default in R.

This dataset contains fuel consumption and design specifications for 32 automobiles. The key variables are:

- **Exposure:** Vehicle weight (wt)
 - **Mediator:** Horsepower (hp)
 - **Outcome:** Fuel efficiency (mpg)
1. Estimate the **Natural Indirect Effect (NIE)** of vehicle weight on fuel efficiency through horsepower.
 - a. Estimate the **NIE** using either the **Product Method** or the **Difference Method**.
 - b. Report the estimated **NIE** and provide a clear interpretation of the results.
 2. Conduct a statistical test to determine whether the indirect effect is significantly different from zero.
 - a. State the null and alternative hypotheses.
 - b. Perform the **Sobel Test** and the **Joint Significance Test** and report the test statistics, p-values, and your conclusions.
- 這學期也即將進入尾聲，誠摯的感謝大家一路以來的配合與投入！為了讓這門課之後能變得更有興趣、更實用、也更貼近大家的需求，想邀請同學們花一點時間分享你們的想法與建議。不論是課程內容、整體安排、上課節奏、難易度、你喜歡或不太喜歡的上課形式，甚至是作業與配分方式，都非常歡迎提出來。沒有標準答案，也不需要客氣，任何真實的回饋都對課程改進非常有幫助。希望能聽到大家的聲音，讓這門課一起變得更加有價值學習。
(不限字數，無話可說也可以不寫)

